3.5.1 is the same with 3.2.4

3.5.2 (a) Prove formula (3.10) $\operatorname{div}(w) = \operatorname{div}(f) + (k/2)\operatorname{div}(d\tau)$.

Pf: $\overset{w=w(f)}{w} \in \Omega^{\otimes k/2}(X(\Gamma))$ $\Rightarrow$ $w = f(\tau)\, b(\tau)^{k/2}$

$\Rightarrow \operatorname{div}(w) = \operatorname{div}(f\tau(d\tau)\cdots(d\tau)) = \operatorname{div}(f) + \frac{k}{2}\operatorname{div}(d\tau)$.

(b) Prove formula (3.11)

$$\lfloor \operatorname{div}(f) \rfloor = \operatorname{div}(w) + \sum_i \lfloor \tfrac{k}{4} \rfloor x_{2,i} + \sum_i \lfloor \tfrac{k}{3} \rfloor x_{3,i} + \sum_i \lfloor \tfrac{k}{2} \rfloor x_i$$

Pf: Formal definition $\operatorname{div}(d\tau) = -\sum_i \frac{1}{2} x_{2,i} - \sum_i \frac{2}{3} x_{3,i} - \sum_i x_i$.

$$\Rightarrow \operatorname{div}(f) = \operatorname{div}(w) - \tfrac{k}{2}\operatorname{div}(d\tau)$$
$$= \operatorname{div}(w) + \sum_i \tfrac{k}{4} x_{2,i} + \sum_i \tfrac{k}{3} x_{3,i} + \sum_i \tfrac{k}{2} x_i$$

(c) Prove for $k \geq 2$ even, $\lfloor k/4 \rfloor \geq (k-2)/4$.

$\lfloor k/3 \rfloor \geq (k-2)/3$.

Pf: Let $k = 2m$. We want to show $\lfloor m/2 \rfloor \geq (m-1)/2$.

$2l \leq m < 2l+2 \Rightarrow 2l \leq m \leq 2l+1 \Rightarrow \begin{cases} \lfloor m/2 \rfloor = l \\ (m-1)/2 \leq l. \end{cases}$

We want to show $\lfloor k/3 \rfloor \geq (k-2)/3$

$3l \leq k < 3l+3 \Rightarrow$ ~~$3l \leq k \leq 3l+2$~~ $\Rightarrow \lfloor k/3 \rfloor = l$

$\begin{cases} l \text{ odd } k = 3l+1 & \begin{cases} \lfloor k/3 \rfloor = l \\ (k-2)/3 = (3l-1)/3 < \lfloor k/3 \rfloor \end{cases} \\ l \text{ even } k = 3l \text{ or } 3l+2 \Rightarrow & \begin{cases} \lfloor k/3 \rfloor = l \\ (k-2)/3 \leq 3l/3 = l. \end{cases} \end{cases}$

(d) Prove (3.12) $\dim S_k(\Gamma) = l(\lfloor \operatorname{div}(f) - \sum_i x_i \rfloor) = l(\lfloor \operatorname{div}(f) \rfloor) - \varepsilon_\infty$

Pf: $g \in S_k(\Gamma) \iff$ ~~$\operatorname{div}(h)+$~~ $\operatorname{div}(f) - \sum_i x_i \geq 0$, x_i cusps of $X(\Gamma)$. $k \geq 4$.

$\dim S_k(\Gamma) = l(\lfloor \operatorname{div}(f) - \sum_i x_i \rfloor) = l(\lfloor \operatorname{div}(f) \rfloor - \sum_i x_i) = l(\lfloor \operatorname{div}(f) \rfloor) - \varepsilon_\infty$

When $k=2$, $\deg \lfloor \operatorname{div}(f) \rfloor < \varepsilon_\infty$ could happen.

3.5.3 i) Prove Thm 3.5.2: $M_0(SL_2(\mathbb{Z})) = \mathbb{C}$.

$M_k(SL_2(\mathbb{Z})) = \{0\}$ $0 \neq k < 4$. even

$S_k(SL_2(\mathbb{Z})) = \{0\}$ $k < 4$ even.

$$M_k(SL_2(\mathbb{Z})) = S_k(SL_2(\mathbb{Z})) \oplus \mathbb{C} E_k \quad k \geq 4 \text{ even}$$

$$\dim S_k(SL_2(\mathbb{Z})) = \begin{cases} \lfloor k/12 \rfloor - 1 & k \equiv 2 \ (12) \\ \lfloor k/12 \rfloor & \text{otherwise} \end{cases}$$

$$M(SL_2(\mathbb{Z})) = \mathbb{C}[E_4, E_6], \quad S(SL_2(\mathbb{Z})) = \Delta \cdot M(SL_2(\mathbb{Z})).$$

Pf: For $k=0$, modular forms are constant functions

For $k<4$ even, space is given as $\dim = 0$.

For $k \geq 4$ even, E_k holomorphic but not cuspidal and any noncuspidal form differs from a multiple of E_k by a cusp form.

Thus $M_k = S_k \oplus \mathbb{C} E_k$.

At i, $\lfloor \frac{k}{4} \rfloor \varepsilon_2 = \lfloor \frac{k}{4} \rfloor$, at μ_3, $\lfloor \frac{k}{3} \rfloor \varepsilon_3 = \lfloor \frac{k}{3} \rfloor$.

Thus $\dim S_k = \lfloor \frac{k}{12} \rfloor - \varepsilon = \lfloor \frac{k}{12} \rfloor - \begin{cases} 1 \\ 0 \end{cases}$

Compare q-expansions: $E_4 = 1 + 240q + \cdots$

$E_6 = 1 - 504q + \cdots$

So no ~~non~~trivial relation.

ii) Write basis for $M_k(SL_2(\mathbb{Z}))$ $\forall k \leq 24$ as polynomials in E_4, E_6.

$E_4^a E_6^b$ $4a+6b=k$, the number equals $\dim M_k$ so they span M_k.

Δ is the unique normalized cusp form of weight 12

Every cusp form vanishes at ∞, hence divisible by Δ

$$\Delta : M_k \xrightarrow{\sim} S_{k+12}$$
$$f \mapsto \Delta \cdot f.$$

3.5.4 (a) Let p be prime. Ex 3.1.4 (b) showed $X_0(p)$ has exactly 2 cusps. Show $\mathcal{M}_2(\Gamma_0(p)) = S_2(\Gamma_0(p)) \oplus \mathbb{C}G_{2,p}$,

$$G_{2,p}(\tau) = G_2(\tau) - p\, G_2(p\tau) \in \mathcal{M}_2(\Gamma_0(p))$$

By dimension formula,

$$\dim \mathcal{M}_2(\Gamma_0(p)) = \dim S_2(\Gamma_0(p)) = \cancel{g_\infty - 2}\ 1 \text{ (not } \varepsilon_\infty).$$

$G_{2,p}$ is modular, in $\mathcal{M}_2(\Gamma_0(p))$, not cusp form.

(We use this $G_{2,p}$ since E_2 not modular)

(b) Let $k, N > 0$, $k(N+1) = 24$, k even.

Thus $(k, N) = (2, 11), (4, 5), (6, 3), (8, 2), (12, 1)$

Show $\dim S_k(\Gamma_0(N)) = 1$.

It follows from P1 p 3.2.2 (b) that

$$S_k(\Gamma_0(N)) = S_k(\Gamma_1(N)) = \mathbb{C}\varphi_k$$

where $\varphi_k(\tau) = \eta(\tau)^k \eta(N\tau)^k$.

Pf: $\underset{\substack{\| \\ 1-g}}{\dim S_k(\Gamma_0(N))} = 1$. by dimension formula.

φ_k has weight k, holomorphic everywhere

vanishes at all cusps

Thus $S_k(\Gamma_0(N)) = \mathbb{C}\varphi_k$.

(c) Combine (a) and (b) to say about $\mathcal{M}_2(\Gamma_0(11))$.

Sol: $\mathcal{M}_2(\Gamma_0(11)) = S_2(\Gamma_0(11)) \oplus \mathbb{C}G_{2,11}$

$$= \mathbb{C}\,\eta(\tau)^2\eta(11\tau)^2 \oplus \mathbb{C}\,G_{2,11}$$

3.5.5 Combine Ex 3.1.4, 3.5.2 to show $k = p+1$, p odd prime

$S_k(SL_2(\mathbb{Z})) \simeq S_2(\Gamma_0(p))$ as $\mathbb{C}$-vector space.

What is the isomorphism when $p = 11$?

Sol: $\dim S_{p+1}(SL_2(\mathbb{Z})) = (p-1)/12$ p odd prime

$\dim S_2(\cancel{SL_2(\mathbb{Z})}\ \Gamma_0(p)) = (p-1)/12$

Define $\Phi(f)(\tau) = f(\tau) \cdot \dfrac{d\tau}{d(p\tau)}$ $X_0(p) \to X(1)$.

$S_{p+1}(SL_2(\mathbb{Z})) \to S_2(\Gamma_0(p))$.

In the case $p = 11$,

$S_{12}(SL_2(\mathbb{Z})) = \mathbb{C}\Delta$, $S_2(\Gamma_0(11)) = \mathbb{C}\eta(\tau)^2\eta(11\tau)^2$

Thus $\Delta \mapsto \eta(\tau)^2\eta(11\tau)^2$

will do.